

A SYNOPSIS ON

TrueCost AI

Submitted in partial fulfilment of the requirement for the award of the degree of

**MASTER OF COMPUTER APPLICATIONS
(MCA)**

Submitted by:

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Under the Guidance of
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CANDIDATE’S DECLARATION

I/we hereby certify that the work which is being presented in the Synopsis entitled “**TrueCost AI**” in partial fulfillment of the requirements for the award of the Degree of Master of Computer Applications (**MCA**) in the Department of Computer Applications of the Graphic Era (Deemed to be University), Dehradun shall be carried out by the undersigned under the supervision of **Dr. Bhawnesh Kumar, Professor**, Department of Computer Applications, Graphic Era (Deemed to be University), Dehradun.

Name: Jatin Kadiyan

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signature

The above mentioned students shall be working under the supervision of the undersigned on
the “**TrueCost AI**”

Signature

Supervisor

Signature

Head of the Department

Internal Evaluation (By Project Committee)

Status of the Synopsis: Accepted / Rejected

Any Comments:

Name of the Committee Members:

Signature with Date

1.

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Chapter 1

Introduction and Problem Statement

1.1 Introduction

In the modern education landscape, higher education has become increasingly expensive, leading to a significant rise in dependence on education loans among students. Financial institutions and digital platforms provide various loan facilities, enabling students to pursue professional degrees in engineering, management, medicine, and international programs. However, the decision to take an education loan is often made without a complete understanding of its long-term financial implications.

Most existing financial tools, such as EMI calculators, focus only on short-term repayment metrics like monthly installment amounts. While EMI provides a basic understanding of repayment, it fails to capture the broader financial impact, including total interest paid, inflation-adjusted cost, opportunity cost of money, and long-term financial constraints.

In addition, students frequently rely on optimistic assumptions regarding future salaries, often influenced by college placement statistics or peer expectations. These assumptions are not always realistic and can lead to financial stress, delayed savings, and reduced financial independence.

To address these issues, the proposed system, **“TrueCost AI: A Financial Decision Support System for Education Loans,”** aims to provide a comprehensive and user-centric platform that enables students to evaluate the real cost of education loans. The system integrates financial calculations with decision-support mechanisms such as risk analysis, loan-to-salary comparison, and opportunity cost estimation.

The application will be developed as a web-based system using modern technologies such as React.js for frontend development, Node.js with Express for backend processing, and MongoDB for data management. The system will present complex financial data in a simplified and visual manner, enabling users to make informed decisions.

1.2 Problem Statement

Despite the availability of financial tools and loan services, students face several challenges while making education loan decisions:

- Existing loan calculators provide only EMI values and do not reflect the total repayment burden.

- Students lack awareness of hidden costs such as interest accumulation over long durations.
- There is no mechanism to compare loan amount with expected future income.
- Placement data provided by institutions is often generalized and may not represent realistic outcomes.
- Users are unable to visualize the long-term financial impact of their decisions.
- Lack of user-friendly tools that convert financial calculations into meaningful insights.

Therefore, there is a need for a system that provides **transparent, comprehensive, and decision-oriented financial analysis**, helping students evaluate whether a loan is financially feasible and sustainable.

Chapter 2

Background/ Literature Survey

In recent years, several tools and systems have been developed to assist individuals in financial planning and loan management. Traditional EMI calculators provided by banks and financial websites are widely used for estimating monthly loan payments. These tools use standard mathematical formulas to calculate EMI based on loan amount, interest rate, and tenure.

However, these systems have several limitations. They do not consider external factors such as inflation, salary growth, or opportunity cost. Furthermore, they are not designed specifically for education loans, where repayment is highly dependent on future income rather than current financial stability.

Research in financial behavior indicates that individuals often underestimate long-term financial obligations due to limited financial literacy. Studies also highlight that high levels of debt can influence career decisions, forcing individuals to prioritize stable income over growth opportunities.

Modern financial platforms have introduced advanced analytics and visualization tools, but they are often complex and not easily understandable for students, especially those from non-technical or rural backgrounds.

Key gaps identified in existing systems:

- Lack of integration between loan calculations and career outcomes
- Absence of risk evaluation mechanisms
- Limited visualization of long-term financial impact
- No focus on student-specific decision-making

The proposed system addresses these gaps by combining financial calculations with intuitive visualization and risk analysis, specifically tailored for education loan scenarios.

Chapter 3

Objectives

1. To design and develop a web-based application for education loan analysis.
2. To calculate EMI, total repayment amount, and total interest using standard financial formulas.
3. To evaluate the feasibility of loan repayment using loan-to-salary ratio.
4. To classify financial risk into categories such as low, medium, and high.
5. To estimate opportunity cost by comparing loan repayment with potential investment returns.
6. To provide visual representations such as graphs and charts for better understanding.
7. To create a user-friendly interface that simplifies complex financial information.
8. To assist students in making informed and rational financial decisions.

Chapter 4

Hardware and Software Requirements

4.1 Hardware Requirements

Sl. No	Name of the Hardware	Specification
1	Computer System	Minimum Intel i5 Processor or equivalent
2	RAM	Minimum 8 GB
3	Storage	Minimum 256 GB SSD
4	Internet Connectivity	Required for development and testing

4.2 Software Requirements

Sl. No	Name of the Software	Specification
1	Operating System	Windows 10/11 or Linux
2	Frontend Technology	React.js
3	Backend Technology	Node.js (Express)
4	Database	MongoDB
5	Development Tools	Visual Studio Code

Chapter 5

Possible Approach/ Algorithms

5.1 System Architecture

The system will follow a three-tier architecture:

- **Presentation Layer (Frontend):** Developed using React.js for user interaction.
- **Application Layer (Backend):** Developed using Node.js and Express for handling logic and calculations.
- **Data Layer (Database):** MongoDB for storing user data and scenarios.

5.2 Working Methodology

The system will operate in the following steps:

1. The user enters input data such as loan amount, interest rate, tenure, and expected salary.
2. The backend processes the input using predefined algorithms.
3. The system calculates EMI, total repayment, and interest.
4. Loan-to-salary ratio is computed to evaluate repayment feasibility.
5. Opportunity cost is estimated based on investment assumptions.
6. The system assigns a risk level based on predefined thresholds.
7. Results are displayed using charts and graphical representations.

5.3 Algorithms Used

(A) EMI Calculation

$$EMI = \frac{P \times R \times (1 + R)^N}{(1 + R)^N - 1}$$

Where:

- P = Loan amount
- R = Monthly interest rate
- N = Number of months

(B) Loan-to-Salary Ratio

$$\text{Loan Ratio} = \frac{\text{Loan Amount}}{\text{Annual Salary}}$$

Risk Classification:

- Low Risk: Ratio ≤ 1
- Medium Risk: $1 < \text{Ratio} \leq 2$
- High Risk: Ratio > 2

(C) Opportunity Cost Calculation

$$\text{Future Value} = \text{EMI} \times r(1+r)^n - 1$$

Where:

- r = Rate of return
- n = Number of months

5.4 Expected Output

The system will generate:

- EMI value
- Total repayment amount
- Total interest payable
- Loan risk category
- Opportunity cost estimation
- Graphical financial insights